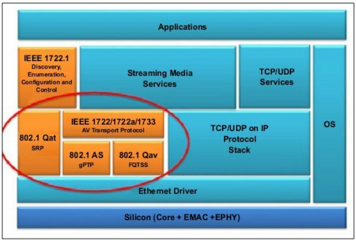

Figure 1: AVB stack layout

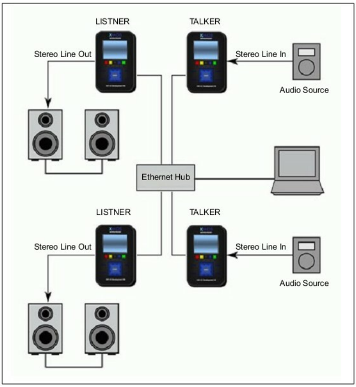

Figure 2: A simple talker and listener set-up

ensure interoperability and improve audio/video quality.

In general, an AVB ecosystem deploys the following protocols:

IEEE 802.1AS: Timing and synchronisation for time-sensitive applications (gPTP)

IEEE 802.1Qav: Forwarding and queuing for timesensitive streams (FQTSS)

IEEE 802.1Qat: Stream reservation protocol (SRP)

IEEE 1722 Layer 2 AVB transport protocol

IEEE 1722.1: Audio video discovery, enumeration, connection management and control protocol (AVDECC)

IEEE 1722.MAAP: MAC address acquisition protocol

How the AVB stack resides in an AVB device can be understood from Figure 1.

The talker and listener set-up forming an AVB network can be understood from Figure 2.

Let me introduce you to the first four basic AVB protocols required to form an AVB network. With these standards, one can have a single talker-listener set-up working in an AVB network. But to establish an AVB network comprising multiple talkers and listeners, one needs to provide the support of the other two (AVDECC and MAAP) standards to manage and control these devices.

IEEE 802.1AS

The IEEE 802.1AS protocol plays a pivotal role in synchronising AVB devices by exchanging timing information over the network. This exchange of timing information allows the AVB devices to synchronise their local clocks with reference to a standard AVB clock.

The model adopted by gPTP is analogous to the slave-master model, where the master exchanges its timing information with the other AVB devices (slaves) and the slaves synchronise their clock with the master (see Figures 3 and 4).

IEEE 802.1Qav

This traffic shaping (queuing and forwarding) protocol is implemented at the hardware layer (MAC). Traffic shaping involves prioritising the transmission of packets to maintain the standard streaming rate required to transfer AVB packets while handling non-AVB network traffic. It can also be termed as bandwidth reservation protocol. To recap, the definition of bandwidth means the number of bits transmitted per unit of time, and the hardware controls the number of packets transmitted per unit of time, using credits.
Let us assume credit to be an X constant value. Now, whenever there is an AVB packet to be transmitted, the hardware checks for the value of X. The credits, i.e., the value of X, keeps getting accumulated at a constant rate when an AVB packet is queued. If the value of X is positive, the AVB packet is transmitted out

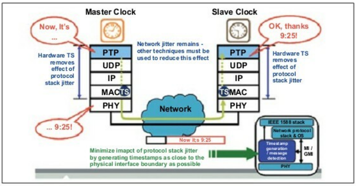

Figure 3: Clock synchronisation using gPTP

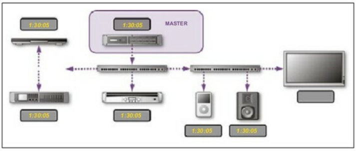

Figure 4: Slaves synchronised to master